

Machine Learning Algorithms

— A Visual Framework

10 foundational algorithms, classified by learning type, application domain, and real-world use case

KARTHIK · PORTFOLIO ARTIFACT
AI / ML / DL / GENAI
REFERENCE SHEET V1.0

How the field nests

AI is the broadest discipline — systems that mimic intelligence. ML is the subset that learns from data. Deep Learning is ML using multi-layer neural networks. Generative AI is the newest layer — DL models that *create* rather than just predict.

- AI — broad: systems that mimic intelligence
- ML — learns patterns from data
- DL — ML using deep neural networks
- GenAI — DL that creates new content

LEARNING TYPE:

Supervised

Unsupervised

Both / Self-Supervised

DOMAIN:

TABULAR

COMPUTER VISION

NLP

GENERATIVE AI

01

SUPERVISED

Linear & Logistic Regression

Fits a straight-line (or sigmoid) relationship between input features and a target. Linear regression predicts continuous values; logistic regression predicts probabilities for classification.

TABULAR

EXAMPLE USE CASES

Demand forecasting, credit scoring, sales lift modeling, churn prediction.

02

SUPERVISED

Decision Trees

Splits data into a tree of if-then rules based on features that best separate the target. Highly interpretable — you can trace every prediction back to a rule path.

TABULAR

EXAMPLE USE CASES

Loan approval logic, supplier risk tiering, customer segmentation rules, medical triage.

03

SUPERVISED

Random Forest

Builds many decision trees on random subsets of data and features, then averages their votes. Reduces overfitting and handles missing values and mixed data types well.

TABULAR

EXAMPLE USE CASES

Fraud detection, inventory stockout risk, sales forecasting baselines, attrition modeling.

04

SUPERVISED

Gradient Boosting (XGBoost, LightGBM)

Builds trees sequentially, each one correcting the errors of the previous. The dominant algorithm for tabular ML competitions and most enterprise prediction tasks today.

TABULAR

EXAMPLE USE CASES

Demand sensing, dynamic pricing, lead scoring, defect prediction, ranking models.

05

SUPERVISED

Support Vector Machines

Finds the optimal boundary (hyperplane) that maximally separates classes. Works well on small, clean, high-dimensional datasets — text classification was a classic SVM stronghold.

TABULAR

NLP

EXAMPLE USE CASES

Document classification, image categorization (pre-deep-learning), bioinformatics, sentiment tagging.

06

UNSUPERVISED

K-Means Clustering

Groups unlabeled data into K clusters by iteratively assigning points to the nearest cluster center, then recomputing centers. The go-to algorithm for fast, exploratory segmentation.

TABULAR

EXAMPLE USE CASES

Customer segmentation, SKU clustering for assortment planning, store grouping, anomaly screening.

07

UNSUPERVISED

Principal Component Analysis

Reduces many correlated features into a few uncorrelated "principal components" that capture most of the variance. Used for compression, visualization, and noise reduction.

TABULAR

COMPUTER VISION

EXAMPLE USE CASES

Visualizing high-dim data, feature engineering, image compression, sensor signal de-noising.

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SUPERVISED · DEEP LEARNING

Convolutional Neural Networks

Neural networks that scan images with small filters to detect edges, textures, and shapes, then combine them into higher-level features. The foundation of modern computer vision.

COMPUTER VISION

EXAMPLE USE CASES

Quality inspection on production lines, shelf-image auditing, medical imaging, autonomous vehicle perception.

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SUPERVISED · DEEP LEARNING

RNNs & LSTMs

Neural networks with memory — they process sequences one step at a time, carrying state forward. LSTMs handle long-range dependencies. Now largely succeeded by Transformers.

TABULAR

NLP

EXAMPLE USE CASES

Time-series forecasting, demand sensing, speech recognition, legacy NLP pipelines.

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SELF-SUPERVISED · DEEP LEARNING

Transformers & LLMs

Use self-attention to weigh every token against every other token in parallel. Pre-trained on huge corpora via self-supervision, then fine-tuned. The engine behind GPT, Claude, and BERT.

NLP

GENERATIVE AI

EXAMPLE USE CASES

Chatbots, code generation, document summarization, RAG search, GenAI copilots.

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SELF-SUPERVISED · GENERATIVE

Diffusion Models

Learn to generate images by reversing a noise process — start from pure noise and iteratively denoise into a coherent image. Powers Stable Diffusion, DALL-E, and Midjourney.

COMPUTER VISION

GENERATIVE AI

EXAMPLE USE CASES

Marketing creative generation, product mock-ups, synthetic training data, image editing & inpainting.

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REINFORCEMENT

Reinforcement Learning

An agent learns by trial and error — taking actions in an environment and receiving rewards. Distinct from supervised/unsupervised: the data is generated by the agent itself.

TABULAR

NLP

EXAMPLE USE CASES

Inventory replenishment policies, route optimization, RLHF for LLM alignment, robotics, game AI.

Picking the right algorithm by domain

Different data shapes call for different model families. Use this as a quick decision aid.

TABULAR DATA

Rows-and-columns data (forecasts, transactions, customers). Start with **Gradient Boosting (XGBoost)** for accuracy or **Logistic Regression** for interpretability. Use **K-Means** for segmentation.

COMPUTER VISION

Images, video, scans. **CNNs** remain the workhorse for classification & detection; **Vision Transformers** are increasingly competitive. **Diffusion** models for generation & synthetic data.

NLP

Text, speech, code. **Transformers** dominate — fine-tune a pre-trained model rather than training from scratch. **SVM/Logistic Regression** still useful for simple, low-latency classification.

GENERATIVE AI

LLMs (Transformers) for text and code; **Diffusion models** for images, video, and audio. Combine with **RLHF** (reinforcement learning) to align outputs with human preferences.